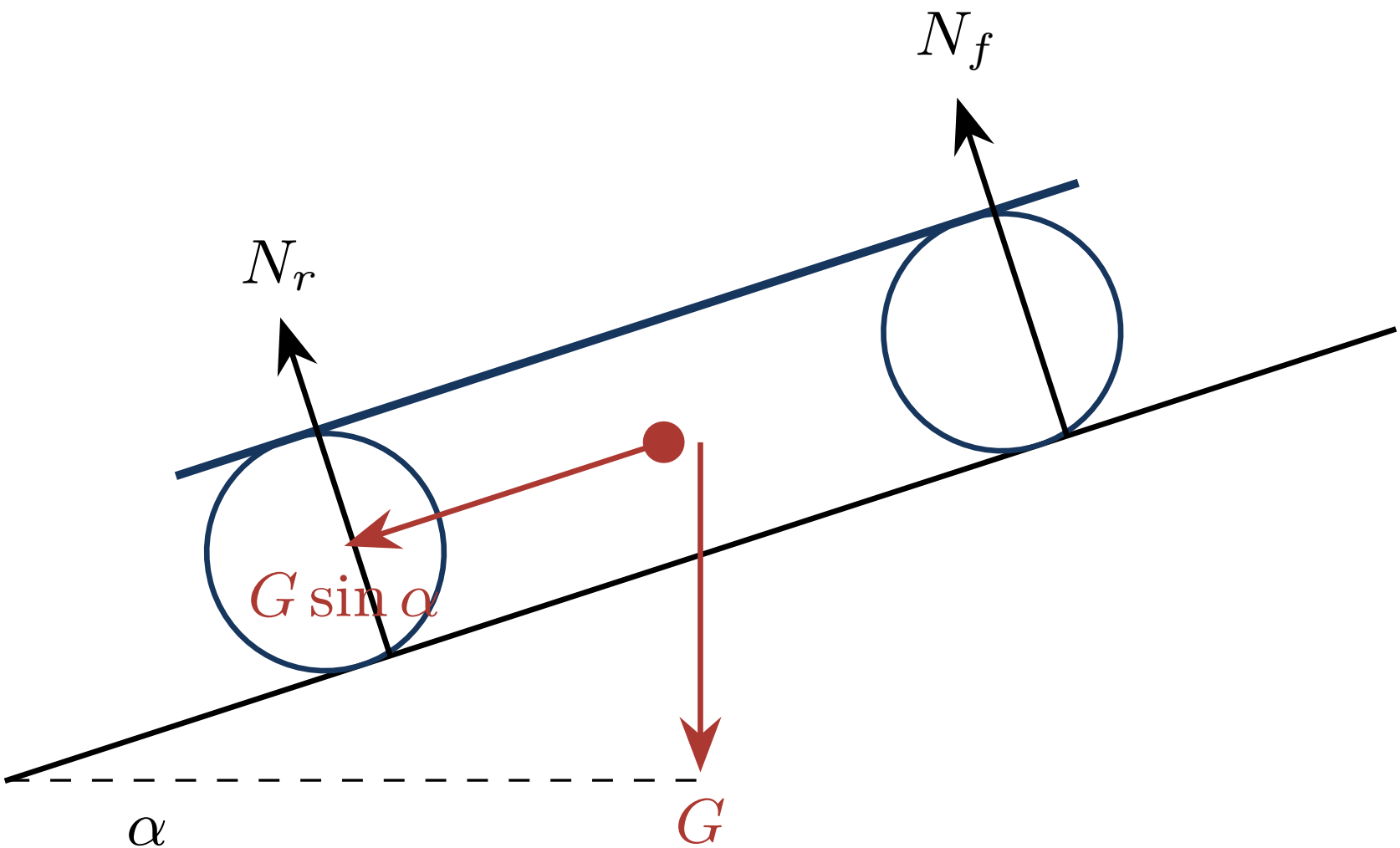$$N_f + N_r = G\cos\alpha$$